

NHS Waiting Times Analysis Dashboard

Project Report

1. Introduction

This project was developed to analyse NHS-style waiting-time data and demonstrate practical skills in data preparation, SQL analysis and Power BI reporting.

The project uses Excel, MySQL and Power BI as part of a complete data analysis workflow.

The dataset contains 5,000 synthetic patient records and was created specifically for learning and portfolio purposes.

2. Project Aim

The main aim was to understand the waiting-time data and identify useful information that could support operational reporting.

The analysis focused on patient waiting times, target breaches, hospitals, specialties, priorities and patient outcomes.

3. Objectives

The project objectives were to:

- Review and clean the dataset.
- Check the quality of the data.
- Prepare the data for SQL analysis.
- Import the data into MySQL.
- Carry out SQL analysis.
- Create a hospital-level summary view.
- Connect MySQL to Power BI.
- Create DAX measures for key performance indicators.
- Develop an interactive dashboard.
- Communicate the main findings clearly.

4. Data Preparation

The dataset was initially reviewed in Excel.

I checked the data structure and reviewed the different fields to make sure they were suitable for analysis.

I also checked dates, duplicate records and data formats before preparing the dataset as a CSV file.

The cleaned CSV was then imported into MySQL.

5. SQL Analysis

MySQL was used to analyse the dataset and answer practical questions about waiting-list performance.

The analysis included:

- Total number of patients
- Average waiting days
- Minimum waiting days
- Maximum waiting days
- Patients by hospital
- Patients by specialty
- Patients by priority
- Patient outcomes
- Target status
- Breached patients
- Breach rate
- Referral trends

A hospital_waiting_summary view was also created.

The view summarises hospital-level performance using measures such as total patients, average waiting days, minimum and maximum waiting times, breached patients and breach rate.

6. Power BI

The MySQL database was connected directly to Power BI Desktop.

The cleaned patient-level table was used as the main source for the dashboard, while the hospital summary view provided additional hospital-level information.

I created DAX measures for the main KPIs, including:

- Total Patients
- Average Waiting Days
- Breached Patients
- Breach Rate
- Patients Still Waiting

These measures allow the dashboard figures to respond to filters selected by the user.

7. Dashboard Design

The dashboard was designed to provide a straightforward overview of waiting-list performance.

KPI cards were used for the main measures, while charts were used to compare hospitals, specialties, priorities and outcomes.

A target-status donut chart was used to show the number of patients within the target compared with those who had breached it.

Additional charts were used to show waiting-time trends and differences between hospitals and specialties.

Slicers were added to allow users to explore the data interactively.

8. Findings

The analysis showed differences in waiting times across hospitals and specialties.

The data also showed that some patients had exceeded the waiting-time target, while others were still waiting for treatment.

There were differences in waiting times between priority groups, which provides an area for further investigation.

The hospital-level analysis also made it possible to compare average waiting times and breach performance.

These findings demonstrate how structured data analysis can help identify patterns in operational performance.

9. Skills Demonstrated

This project allowed me to demonstrate practical skills in:

Excel

- Data checking
- Data cleaning
- Data preparation

SQL

- SELECT
- COUNT
- AVG
- MIN
- MAX
- SUM
- CASE
- GROUP BY
- ORDER BY

- Date functions
- SQL views

Power BI

- Data connection
- Data modelling
- DAX
- KPI cards
- Charts
- Slicers
- Dashboard design

10. Project Workflow

The complete workflow was:

Excel → CSV → MySQL → SQL Analysis → SQL View → Power BI → DAX → Dashboard

This approach allowed me to demonstrate the complete process rather than focusing only on the final dashboard.

11. Conclusion

This project strengthened my practical understanding of data analysis and reporting.

I was able to take a dataset from its initial preparation, analyse it using SQL and then present the results through an interactive Power BI dashboard.

The project has helped me develop skills that are relevant to junior Data Analyst and Information Analyst positions, particularly in data preparation, SQL, reporting, visualisation and communicating findings.

12. Future Improvements

If this were developed further, I would consider adding:

- More years of data
- More NHS regions
- Additional hospital information
- A separate date dimension
- More detailed waiting-list categories
- Additional Power BI drill-through pages
- Automated data refresh
- Further data-quality checks

13. Data Disclaimer

This project uses synthetic data created for educational and portfolio purposes.

It does not contain real NHS patient information, and the results should not be interpreted as actual NHS waiting-time or hospital performance figures.